

Safety Data Sheet according to UN-GHS

1. Identification

Product identifier:	MERACRYL® GMAA
Chemical name:	Methacrylic acid
Other means of identification	
Recommended use:	Monomer For use in industrial installations or professional treatment only. Chemical Intermediate Consumers
Recommended restrictions:	Applications where liquid monomer is intended to come into contact with skin or nails.

Manufacturer/Importer/Distributor Information

Company Name	:	Roehm Hong Kong Co., Ltd. Unit 1504-05A, 15/F, 909 Cheung Sha Wan Road Cheung Sha Wan, Kowloon Hong Kong
Telephone	:	+86 21 6759 1000
E-mail	:	SDS-PS-Asia@roehm.com

Emergency telephone number:

24-Hour Health Emergency	:	+86 532 8388 9090 (China mainland only) +49 6241 402 5280 (24h)
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2. Hazard(s) identification

Classification according to GHS

Physical Hazards

Flammable liquids	Category 4
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Health Hazards

Acute toxicity (Oral)	Category 4
Acute toxicity (Inhalativ)	Category 4
Acute toxicity (Dermal)	Category 3
Skin corrosion	Sub-category 1A
Serious Eye Damage/Eye Irritation	Category 1
Specific Target Organ Toxicity - Single Exposure	Category 3 (Respiratory system)

Environmental Hazards

Acute hazards to the aquatic environment	Category 3
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Label Elements**Hazard Symbol:****Signal Word:** Danger

Hazard Statement: Combustible liquid.
Harmful if swallowed or if inhaled.
Toxic in contact with skin.
Causes severe skin burns and eye damage.
May cause respiratory irritation.
Harmful to aquatic life.

Precautionary Statements

Prevention: Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. Avoid breathing dust/fume/gas/mist/vapors/spray. Wash thoroughly after handling. Do not eat, drink or smoke when using this product. Use only outdoors or in a well-ventilated area. Avoid release to the environment. Wear protective gloves/protective clothing/eye protection/face protection.

Response: IF SWALLOWED: Rinse mouth. Do NOT induce vomiting. IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water [or shower]. IF INHALED: Remove person to fresh air and keep comfortable for breathing. IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a POISON CENTER or doctor/ physician. Take off immediately all contaminated clothing and wash it before reuse. In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish.

Storage: Store in a well-ventilated place. Keep container tightly closed. Store in a well-ventilated place. Keep cool. Store locked up.

Disposal: Dispose of contents/ container to an approved waste disposal plant.

Other hazards: Polymerization with heat evolution may occur in the presence of radical forming substances (e.g. peroxides), reducing substances, and/or heavy metal ions.

3. Composition/information on ingredients**Substances**

Chemical Identity	Common name and synonyms	CAS number	Content in percent (%)*
methacrylic acid	2-methylprop-2-enoic acid	79-41-4	>99,5%

* All concentrations are percent by weight unless ingredient is a gas. Gas concentrations are in percent by volume.

The exact concentration has been withheld as a trade secret.

4. First-aid measures

Description of first aid measures

General information:

First aider needs to protect himself. Take off all contaminated clothing immediately. Medical treatment is necessary if symptoms occur which are obviously caused by skin or eye contact with the product or by inhalation of its vapours.

Inhalation:

Move subject to fresh air and keep him calm. Call a physician immediately.

Skin Contact:

In case of contact with the skin wash off immediately with plenty of water. Call a physician immediately.

Eye contact:

Keeping the eyelids apart flush thoroughly with water immediately. Protect unharmed eye. Call ambulance. (Cue: caustic burn of the eyes) Continue rinsing eye until arrival at ophthalmic hospital.

Ingestion:

Rinse out mouth and give plenty of water to drink. Do not induce vomiting. Call a physician immediately.

Personal Protection for First-aid Responders:

No data available.

Most important symptoms and effects, both acute and delayed Symptoms:

Acute dermal irritation/corrosion
Acute eye irritation/corrosion
Risk of serious damage to eyes.
Difficulty in breathing, burning in mouth and throat
Coughing.

Hazards:

Strongly irritating to corrosive. Harmful if inhaled. Toxic in contact with skin.

Indication of immediate medical attention and special treatment needed**Treatment:**

Treat symptomatically.

5. Fire-fighting measures

General Fire Hazards:

Standard procedure for chemical fires. Keep unauthorized personnel away. Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. Vapours are heavier than air and can form an explosive mixture with air. Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. Prevent fire extinguishing water from contaminating surface water or the ground water system.

Suitable (and unsuitable) extinguishing media**Suitable extinguishing media:**

Extinguish with foam, carbon dioxide or dry powder.

Unsuitable extinguishing media:

High volume water jet

Special hazards arising from the substance or mixture:

Closed container may rupture if strongly heated. May be released in case of fire: carbon monoxide, carbon dioxide, organic products of decomposition.

Special protective equipment and precautions for fire-fighters**Special fire-fighting procedures:**

Keep away from heat/sparks/open flames/hot surfaces. No smoking. Take action to prevent static discharges. In the event of fire, cool the endangered containers with water. When heated above the flash point and/or during spraying (atomizing), ignitable mixtures may form in air.

Special protective equipment for fire-fighters:

Wear self-contained breathing apparatus.

6. Accidental release measures**Personal precautions, protective equipment and emergency procedures:**

Assure sufficient ventilation. Wear protective gloves/protective clothing/eye protection/face protection. Keep away sources of ignition. Use breathing apparatus if exposed to vapours/dust/mist/aerosol. Do not breathe vapours or spray mist. Do not eat, drink or smoke when using this product. Wash hands thoroughly with soap and water after handling.

Accidental release measures:

No data available.

For emergency responders:**Methods and material for containment and cleaning up:**

Do not inhale vapours / aerosols. Avoid contact with eyes, skin, and clothing. Observe regulations on prevention of water pollution (check, dam up, cover up). Larger quantities: Remove mechanically (by pumping). Use explosion-proof equipment! Smaller quantities and/or residues: Contain with absorbent material (e.g. sand, diatomaceous earth, acid absorbent, universal absorbent or sawdust). Dispose of in accordance with regulations.

Environmental Precautions:

Avoid release to the environment. Prevent product from getting into drains/surface water/groundwater.

7. Handling and storage**Handling****Technical measures:**

Install appropriate equipment and wear appropriate personal protective equipment (see "8. Exposure control/personal protection").

Local/Total ventilation:

Use explosion-proof electrical, ventilating and lighting equipment.

Safe handling advice:

Keep container tightly closed. Provide good room ventilation even at ground level (vapours are heavier than air). Methacrylic acid freezes at 15°C (59°F). Improper thawing can result in violent polymerization. The thawing of frozen drums can best be accomplished by placing them in a heated room at temperatures up to 40°C (104°F). This will allow the acid to melt slowly within a 48-hour period. DO NOT remove any material if stock is frozen or partially

frozen. Mix during and after thawing to properly distribute inhibitor. NEVER use steam or electric heating bands. Do not breathe spray. Do not breathe vapors. Absolutely avoid contact with the eyes and/or skin. Wash thoroughly after handling. Do not eat, drink or smoke during use. Keep away from heat/sparks/open flames/hot surfaces. No smoking. Take action to prevent static discharges. In the event of fire, cool the endangered containers with water. When heated above the flash point and/or during spraying (atomizing), ignitable mixtures may form in air. Follow all SDS/label precautions even after container is emptied because it may retain product residues. The product should only be handled by trained personnel. Safety shower and eye wash fountain should be available.

Contact avoidance measures:

see section 8. see section 10.

Storage**Safe storage conditions:**

Fill the container by approximately 90 % only as oxygen (air) is required for stabilisation. With large storage containers make sure the oxygen (air) supply is sufficient to ensure stability. Maintain the temperature of the methacrylic acid between 18°C and approx. 35°C. Protect from light. The ideal storage temperature is 20-25°C. Can polymerize with intense heat release. Higher temperature such as 40° C can be safely tolerated for short times such as during transportation or inventory turnover.

Safe packaging materials:

No data available.

8. Exposure controls/personal protection

Control Parameters**Occupational Exposure Limits**

Observe national threshold limit values.

Biological Limit Values

No biological exposure limits noted for the ingredient(s).

Appropriate Engineering Controls

For monitoring procedures refer for instance to "Empfohlene Analysenverfahren für Arbeitsplatzmessungen", Schriftenreihe der Bundesanstalt für Arbeitsschutz and "NIOSH Manual of Analytical Methods", National Institute for Occupational Safety and Health Eye wash facilities and emergency shower must be available when handling this product.

Individual protection measures, such as personal protective equipment**General information:**

No data available.

Eye/face protection:

tightly fitting goggles

Hand Protection:

Material: butyl rubber gloves (minimal thickness 0.3 mm)
Break-through time: 480 min
Guideline: EN 374
Additional Information: nitrile rubber gloves, Suitable as spray protection.
Additional Information: Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. Also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion, and the contact time., The above mentioned hand protection is based on special knowledge of the chemical and the intended handling of this product, however, it still may not be suited for all workplaces. A qualified hazard assessment should be made prior to the onset of work in order to determine the suitability of gloves for specific working environments and processes., Gloves should be discarded and replaced if there is any indication of degradation or chemical breakthrough.

Other:

On handling of larger quantities: face mask, chemical-resistant boots and apron

Respiratory Protection:

Breathing apparatus in case of high concentrations

Hygiene measures:

Store work clothing separately. Take off all contaminated clothing immediately. Follow the usual good standards of occupational hygiene. Clean skin thoroughly after work; apply skin cream.

9. Physical and chemical properties**Information on basic physical and chemical properties****Appearance**

Physical state: liquid
Form: liquid
Color: Colorless

Odor: stinging

Odor Threshold: No data available.

Freezing point: 59,9 °F/15,5 °C

Boiling Point: 324 °F/162 °C (1.013 hPa)

Flammability: Combustible liquid. Not classified as flammable but will burn.

Upper/lower limit on flammability or explosive limits

Explosive limit - upper: No data available.

Explosive limit - lower: 2,1 %(V)

Flash Point: 153 °F/67 °C

Autoignition Temperature: The substance or mixture is not classified as pyrophoric. 752 °F/400 °C Auto Ignition Temperature

Decomposition Temperature: Not applicable

pH: 2 - 2,2 (100 g/l, 77 °F/25 °C)

Viscosity

Dynamic viscosity: 1,38 mPa.s (68 °F/20 °C, Brookfield)

Kinematic viscosity: 1,36 mm²/s (77 °F/25 °C) (literature value)

Flow Time: No data available.

Solubility(ies)

Solubility in Water: 98 g/l (68 °F/20 °C)

Solubility (other): miscible with esters and ketones miscible with alcohols

Partition coefficient (n-octanol/water): 0,93 (Measured)

Vapor pressure: 0,97 hPa (68 °F/20 °C)

Relative density: No data available.

Density: 1,01 g/cm³ (68 °F/20 °C)

Bulk density:

Relative vapor density: > 1 68 °F/20 °C

Other information

Explosive properties: Vapours may form explosive mixtures with air

10. Stability and reactivity

Reactivity: Polymerisation can occur.

Chemical Stability: No decomposition if used as directed.

Possibility of hazardous reactions: Polymerization with heat evolution may occur in the presence of radical forming substances (e.g. peroxides), reducing substances, and/or heavy metal ions. Product reacts violently to explosively with alkali metals, alkaline earth metals, various metal powders, strong alkalis and ammonia. Reactions with alkalis. Reacts with water. Contact with metals and water liberates hydrogen. Reactions with organic substances.

Conditions to avoid: The product is normally supplied in a stabilized form. If the permissible storage period and/or storage temperature is exceeded, the product may polymerize with heat evolution.

Incompatible Materials: Reactions with strong oxidizing agents.

Hazardous Decomposition Products: None when used as directed.

11. Toxicological information

General information: The substance is rapidly metabolized

Information on toxicological effects

Inhalation: Harmful if inhaled.

Skin Contact: Toxic in contact with skin.

Eye contact: Causes eye burns.

Ingestion: Harmful if swallowed.

Symptoms related to the physical, chemical and toxicological characteristics

Inhalation: General respiratory distress, unproductive cough.

Skin Contact: Causes severe skin burns.

Eye contact: Causes eye burns.

Ingestion: Ingestion may cause severe irritation of the mouth, the esophagus and the gastrointestinal tract.

Information on likely routes of exposure

Acute toxicity (list all possible routes of exposure)

Oral

Product: Acute oral toxicity category 4 (UN-GHS)

Components:
methacrylic acid LD 50 (Rat): 1.320 mg/kg

Dermal

Product: Acute dermal toxicity category 3 (UN-GHS)

Components:
methacrylic acid LD 50 (Rabbit): 500 mg/kg

Inhalation

Product: Acute inhalation toxicity category 4 (UN-GHS)

Components:
methacrylic acid LC 50 (Rat, 4 h): 7,1 mg/l Aerosols; moderately toxic after single exposure; Dust and mist Not toxic after single exposure; Vapour

Repeated dose toxicity

Product: NOAEL (Rat(male and female), Inhalativ, 5 days/weeks, 6 hours/day): 100 ppm Findings: damage to the nasal mucosa

Components:
methacrylic acid NOAEL (Rat(male and female), Inhalativ, 5 days/weeks, 6 hours/day): 100 ppm Findings: damage to the nasal mucosa

Skin Corrosion/Irritation

Product: Skin corrosion/irritation - Category 1A

Components:
methacrylic acid OECD Test Guideline 404 (Rabbit): Corrosive. , < 3 min

Serious Eye Damage/Eye Irritation

Product: , Causes serious eye damage.

Components:
methacrylic acid Risk of serious damage to eyes. OECD Test Guideline 405 , Rabbit: Corrosive.

Respiratory or Skin Sensitization

Product: Not classified for respiratory sensitization Not a skin sensitizer.

Components:
methacrylic acid in vivo, OECD Test Guideline 406 (Guinea Pig): Not a skin sensitizer.
Not classified for respiratory sensitization

Carcinogenicity

Product: An Expert Judgment stated that no classification is necessary based on present knowledge.

Components:
methacrylic acid Not classified

Germ Cell Mutagenicity
Not classified

In vitro

Product: Based on available data, the classification criteria are not met.;

Components:
methacrylic acid Bacterial reverse mutation assay (OECD 471): negative
Genetic mutation in mammal cells (OECD 476): negative , Chinese hamster lung fibroblasts (V79)
Micronucleus test (OECD 487): negative

In vivo

Product: No test results available.

Components:
methacrylic acid No test results available.

Reproductive toxicity

Product: An Expert Judgment stated that no classification is necessary based on present knowledge.

Components:
methacrylic acid Not classified

Specific Target Organ Toxicity - Single Exposure

Product: Respiratory tract irritation. Specific target organ toxicity – single exposure Category 3 (UN-GHS)

Components:
methacrylic acid Category 3 with respiratory tract irritation.

Specific Target Organ Toxicity - Repeated Exposure

Product: Based on available data, the classification criteria are not met.

Components:
methacrylic acid Not classified

Aspiration Hazard

Product: No aspiration toxicity classification no evidence for hazardous properties (structure-activity-relationships) (analogy)

Components:
methacrylic acid Not classified

Information on health hazards

Other hazards

Product: Carefully avoid contact with skin and eyes as well as inhalation of product vapours.;

12. Ecological information

Ecotoxicity:

Acute hazards to the aquatic environment:

Fish

Product: LC 50 (Oncorhynchus mykiss, 96 h): 85 mg/l

Components:

methacrylic acid LC 50 (Oncorhynchus mykiss, 96 h): 85 mg/l

Aquatic Invertebrates

Product: EC 50 (Daphnia magna, 48 h): > 130 mg/l (own study)

Components:
 methacrylic acid EC 50 (Daphnia magna, 48 h): > 130 mg/l (own study)

Toxicity to Aquatic Plants

Product: EC 50 (Selenastrum capricornutum (green algae), 72 h): 45 mg/l (OECD 201)

Components:
 methacrylic acid EC 50 (Selenastrum capricornutum (green algae), 72 h): 45 mg/l (OECD 201)

Toxicity to microorganisms

Product: EC10 (Pseudomonas putida, 16,5 h): 100 mg/l ((DIN 38412, part 8)) (own study)

Components:
 methacrylic acid EC10 (Pseudomonas putida, 16,5 h): 100 mg/l ((DIN 38412, part 8)) (own study)

Chronic hazards to the aquatic environment:

Fish

Product: NOEC (Danio rerio (zebra fish), 35 d): 10 mg/l (OECD 210)

Components:
 methacrylic acid NOEC (Danio rerio (zebra fish), 35 d): 10 mg/l (OECD Test Guideline 210)

Aquatic Invertebrates

Product: NOEC (Daphnia magna, 21 d): 53 mg/l (OECD Test Guideline 202)

Components:
 methacrylic acid NOEC (Daphnia magna, 21 d): 53 mg/l (OECD Test Guideline 202)

Toxicity to Aquatic Plants

Product: NOEC (Selenastrum capricornutum (green algae), 72 h): 8,2 mg/l (OECD 201)

Components:
 methacrylic acid NOEC (Selenastrum capricornutum (green algae), 72 h): 8,2 mg/l (OECD 201)

Toxicity to microorganisms

Product: EC10 (Pseudomonas putida, 16,5 h): 100 mg/l ((DIN 38412, part 8)) (own study)

Components:
 methacrylic acid EC10 (Pseudomonas putida, 16,5 h): 100 mg/l ((DIN 38412, part 8)) (own study)

Persistence and Degradability

Biodegradation

Product: 86 % (28 d, OECD 301 D), Readily biodegradable

Components:
 methacrylic acid 86 % (28 d, OECD Test Guideline 301D) The 10 day time window criterion is not fulfilled., The product is biodegradable.

BOD/COD Ratio

Product: No data available.

Components:
 methacrylic acid No data available.

Bioaccumulative potential**Bioconcentration Factor (BCF)**

Product: Accumulation in organisms is not expected due to the coefficient of distribution of n-octanol in water (log Pow).

Components:
methacrylic acid Accumulation in organisms is not expected due to the coefficient of distribution of n-octanol in water (log Pow).

Partition Coefficient n-octanol / water (log Kow)

Product: Log Kow: 0,93 22 °C (Measured)

Components:
methacrylic acid Log Kow: 0,93 22 °C

Mobility in soil:

Product Very sparingly volatile from the aqueous phase. Binding to the solid soil phase, sediment or clarification sludge is not expected. The substance is distributed mainly into the water phase and the soil.

Components:
methacrylic acid Very sparingly volatile from the aqueous phase. Binding to the solid soil phase, sediment or clarification sludge is not expected. The substance is distributed mainly into the water phase and the soil.

Results of PBT and vPvB assessment:

Product Not a PBT, vPvB substance as per the criteria of the REACH Ordinance.

Components:
methacrylic acid No data available.

Other adverse effects:**Other hazards**

Product: Prevent substance from entering soil, natural bodies of water and sewer systems. Photochemical degradation (air) takes place.

13. Disposal considerations

General information: Dispose of waste and residues in accordance with local authority requirements.

Disposal methods: Waste is hazardous. It must be disposed of in accordance with the regulations after consultation of the competent local authorities and the disposal company in a suitable and licensed facility. Strictly controlled conditions during disposal or treatment to air, wastewater and waste. Do not add wastewater to a biological wastewater treatment plant Bring wastewater containing AOX for professional disposal. The waste key number must be determined as per the European Waste Types List (decision on EU Waste Types List 2000/532/EC) in cooperation with the disposal firm / producing firm / official authority.

Contaminated Packaging: Contaminated packaging should ideally be emptied; it can then be recycled after having been decontaminated. Packaging that cannot be cleaned should be disposed of professionally. Uncontaminated packaging may be taken for recycling.

14. Transport information**International Regulations****IATA-DGR**

UN/ID No. : UN 2531
Proper shipping name : Methacrylic acid, stabilized
Class : 8
Packing group : II
Labels : 8
Packing instruction (cargo aircraft) : 855
Packing instruction (passenger aircraft) : 851

IMDG-Code

UN number : UN 2531
Proper shipping name : METHACRYLIC ACID, STABILIZED

Class : 8
Packing group : II
Labels : 8
EmS Code : F-A, S-B
Marine pollutant : no
Remarks : Clear of living quarters., Protected from sources of heat.

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Not applicable for product as supplied.

Special precautions for user

The transport classification(s) provided herein are for informational purposes only, and solely based upon the properties of the unpackaged material as it is described within this Safety Data Sheet. Transportation classifications may vary by mode of transportation, package sizes, and variations in regional or country regulations.

15. Regulatory information**International regulations****Montreal protocol**

Not applicable

Stockholm convention

Not applicable

Rotterdam convention

Not applicable

Kyoto protocol

Not applicable

16. Other information, including date of preparation or last revision

Issue Date: 20.03.2019

Version #: 2.0

Source of information:

relevant manuals and publications
own examinations
own toxicological and ecotoxicological studies
toxicological and ecotoxicological studies of other manufacturers
SIAR
OECD-SIDS
RTK public files

Further Information:

The product is normally supplied in a stabilized form. If the permissible storage period and/or storage temperature is exceeded, the product may polymerize with heat evolution. Caution: In case of thawing crystallised methacrylic acid do not use temperatures above 35 °C. Strictly follow instructions of supplier.

Disclaimer:

This information and all further technical advice is based on our present knowledge and experience. However, it implies no liability or other legal responsibility on our part, including with regard to existing third party intellectual property rights, especially patent rights. In particular, no warranty, whether express or implied, or guarantee of product properties in the legal sense is intended or implied. We reserve the right to make any changes according to technological progress or further developments. The customer is not released from the obligation to conduct careful inspection and testing of incoming goods. Performance of the product described herein should be verified by testing, which should be carried out only by qualified experts in the sole responsibility of a customer. Reference to trade names used by other companies is neither a recommendation, nor does it imply that similar products could not be used.